



WORMLIKE AMPHIBIANS

With their long bodies, body rings, and lack of limbs, caecilians are often mistaken for large earthworms. Most are found in loose soil or deep leaf litter in tropical forests.

also have an extremely wide mouth, which enables them to consume relatively large prey.

Although most adult amphibians breathe using lungs, they do not have a rib cage and a diaphragm to inflate them. Many amphibians can absorb oxygen into their blood through other organs and tissues, including gills, the skin, and the lining of the mouth. The relative importance of each route varies among different life stages and adult forms. Young amphibian larvae breathe using external gills, as do those salamanders that retain the larval form into the adult stage. Most terrestrial frogs and toads, and some salamanders, have lungs, but also absorb a certain amount of oxygen through the skin and the mouth (see panel, opposite). One group of salamanders, the plethodontids, have no lungs and rely entirely on the skin and the mouth to take in oxygen.

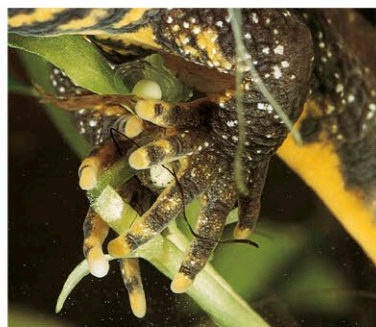
Life cycles

An amphibian's life cycle involves 3 stages—egg, larva, and adult, with the transition from larva to adult involving a radical transformation known as metamorphosis.

Most amphibians deposit their eggs in water. However, some species lay their eggs on land, and others show a variety of ways of retaining them inside their body. In nearly all frogs and toads, fertilization is external, with sperm entering the eggs outside the female's body. During mating, the male clasps the female, usually from above, and sheds sperm into the

water at the same moment as the female expels her eggs. In most newts and salamanders, fertilization is internal, the sperm being transferred to the female in small, gelatinous capsules called spermatophores. This requires precise coordination between the movements of male and female, and many newts and salamanders have elaborate courtship behavior to achieve this. In caecilians, the male has a penislike organ, which he uses to insert his sperm directly into the female.

The eggs of amphibians have no shell, but are enclosed in a gelatinous envelope that quickly shrivels up in dry conditions. Eggs may be laid singly, in strings, or in clumps; they may be carefully attached to plants or other solid objects, or just left to drift in the water. Several amphibians show parental care, guarding their eggs against predators and infection from fungal diseases. The number of eggs laid varies



PARENTAL CARE

European newts lay their eggs individually. Wrapping them in leaves may protect them from UV radiation.

TAKING TO LAND

Swarms of newly metamorphosed European common frogs gather by the side of a pond before leaving the water for the first time.



enormously: some amphibians lay only one or 2 eggs at a time; others lay up to 50,000. The eggs hatch into minute larvae, which retain sufficient yolk to sustain them for a few days.

The larvae of amphibians (called tadpoles in frogs and toads) are very different in appearance, diet, and lifestyle from adults. Metamorphosis involves a major reconstruction of all parts of the body (see panel, below). The larvae of newts and salamanders are carnivorous, feeding on tiny invertebrates. In contrast, the tadpoles of frogs and toads are typically herbivores, either filter feeding on microscopic plants that live suspended in water, or grazing on algae that cover plants, rocks, and other submerged objects.

The larval stage is an important growth phase for amphibians, but different species vary considerably in the amount of total lifetime growth that is completed in the larval rather

than the adult stage. Nearly all amphibians actually become smaller during metamorphosis.

Some amphibians show direct development, in which the larva is not free-swimming but completes its development while still inside the egg and emerges as a tiny version of the adult form. Direct development occurs in a number of species that lay their eggs on land, and also in some that retain their eggs inside the body. Such species, by having no free-swimming larval stage, are not dependent on standing water.

Breeding migrations

Most amphibians live on land as adults but return to water each year to breed. Breeding is usually stimulated by a combination of environmental factors, including temperature, rainfall, and day length. In some species, referred to as explosive breeders, there is a mass migration of adults, all moving toward a breeding pond. Evidence shows that various species locate the same breeding site year after year by using their sense of smell, an ability to navigate by the sun, or by detecting tiny variations in the earth's magnetic field.



MIGRATING FROGS

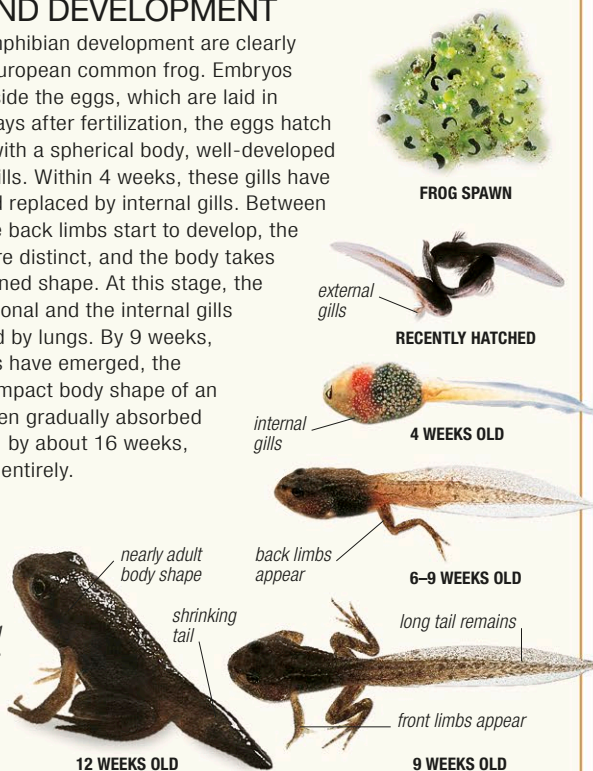
Some species of amphibians, such as these European common frogs, make annual mass migrations to breeding sites, often from great distances and across rough terrain.

GROWTH AND DEVELOPMENT

The 3 stages of amphibian development are clearly illustrated by the European common frog. Embryos start to develop inside the eggs, which are laid in clumps. About 6 days after fertilization, the eggs hatch into tiny tadpoles with a spherical body, well-developed tail, and external gills. Within 4 weeks, these gills have been absorbed and replaced by internal gills. Between 6 and 9 weeks, the back limbs start to develop, the head becomes more distinct, and the body takes on a more streamlined shape. At this stage, the legs are fully functional and the internal gills have been replaced by lungs. By 9 weeks, when the front legs have emerged, the tadpole has the compact body shape of an adult. The tail is then gradually absorbed into the body, until, by about 16 weeks, it has disappeared entirely.

EGG TO ADULT

The European common frog's development, from egg to adult, typically takes about 16 weeks, but varies according to temperature levels and food supply.



KEEPING GUARD

This marbled salamander guards its clutch of eggs at the edge of a pond. The female curls her body around her eggs to protect them from predators and fungal infection.